Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Further Pure 1 – Conic Sections

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Write the standard equation of a parabola with focus $(a, 0)$ and directrix $x = -a$. **[2]**

2. Find the focus and directrix of $y^2 = 8x$. **[3]**

3. Write the standard equation of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. State the foci. **[3]**

4. Find the eccentricity of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$. **[3]**

5. Write the standard equation of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. **[2]**

6. Find the asymptotes of $\frac{x^2}{9} - \frac{y^2}{4} = 1$. **[3]**

7. Parametric equations of a parabola: $x = at^2$, $y = 2at$. Find the equation of the tangent at t. **[4]**

8. Find the equation of the normal to $y^2 = 4ax$ at the point $(at^2, 2at)$. **[4]**

9. For the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, write parametric equations. **[2]**

10. Find the equation of the tangent to the ellipse at $(a\cos\theta, b\sin\theta)$. **[4]**

11. Show that the eccentricity e of an ellipse satisfies $b^2 = a^2(1 - e^2)$. **[3]**

12. Find the focus of the parabola $x^2 = 12y$. **[3]**

13. Identify the conic $x^2 + 4y^2 = 16$. **[2]**

14. Identify the conic $xy = 4$. **[2]**

15. Find the vertices of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$. **[2]**

Section B – Hard (Q16–25)

16. Find the equation of the chord of the parabola $y^2 = 4ax$ joining t_1 and t_2. **[4]**

17. Show that the condition for $y = mx + c$ to be tangent to $y^2 = 4ax$ is $c = \frac{a}{m}$. **[4]**

18. Find the locus of the midpoint of a focal chord of $y^2 = 4ax$. **[5]**

19. For the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, find the equation of the normal at $(a\cos\theta, b\sin\theta)$. **[4]**

20. Find the eccentricity and foci of $\frac{x^2}{16} - \frac{y^2}{9} = 1$. **[4]**

21. Show that the pair of tangents from (x_1, y_1) to the circle $x^2 + y^2 = a^2$ have equation $T^2 = SS_1$. **[4]**

22. Find the condition that the line $y = mx + c$ touches the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. **[4]**

23. Parametric point $P(at^2, 2at)$ on the parabola. Find the area of the triangle formed by the tangents at $t = 1$ and $t = 2$ and the x-axis. **[5]**

24. Reduce $9x^2 + 4y^2 - 18x + 16y - 11 = 0$ to standard form and identify the conic. **[5]**

25. Find the director circle of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. **[3]**

End of Worksheet • *Show all working clearly.*